

COGS17 Neurobiology of Cognition
Lecture 11: Learning and Memory

LEARNING = The development of a permanent change in behavior based on experience

- **Law of Effect**: Stimulus/Context/Act associated with Reinforcement, will be repeated = “Conditioning”
 - Classical Conditioning = Developed association between stimuli
 - e.g. Dog salivates to Food. Pair Food (Pos Reinforcer=StimA) w/Bell (Conditioned Stimulus=StimB);
Soon dog salivates to Bell alone, via association between Stim A & Stim B
- **Temporal Contiguity** critical in both – i.e. above events must Co-Occur to become associated in learner’s mind
 - Presumably, it is this co-occurrence that leads to the neural co-activity of the stimulated circuits
- **Hebbian Cell Assemblies** (proposed by Hebb as the fundamental neural process involved in Learning)
 - = Co-activated neural circuits involved in learning and retrieval of associations (Fire Together:Wire Together)
 - Mechanisms responsible include structural or metabolic changes in NT availability, release, and/or reception
 - In contemporary computational models, involves development of “weighting” changes across Neural Nets

Long-Term Potentiation (LTP) (Semi-)permanent structural and connectivity changes, via variety of mechanisms:

- **Post-Synaptic**: Include structural changes in Receptor Sites, Dendritization, etc.
 - e.g. **Hippocampus** in the Rat, in which the best-studied mechanisms of LTP involve...
 - Cells w/multiple types of Glutamate Receptor Sites on the same cell; Some sites harder to trigger than others
 - **AMPA** Receptor Sites respond well to Glutamate (Ionotropic: Allow Na⁺ in to excite Post-Synaptic cell)
 - **NMDA** Receptor Sites only respond to Glutamate if cell already partially hypo-polarized by above
 - Typically Magnesium ions (**Mg⁺⁺**) block these gates that, when opened, allow Na⁺ & Ca⁺⁺ to enter
 - When Post-Synaptic cell is massively stimulated by Glutamate from multiple Pre-Synaptic cells...
 - AMPA receptors reduce polarity of cell, evict Mg⁺⁺, allowing NMDA Receptors to also respond
 - Ca⁺⁺ influx helps change structure of Post-Synaptic cell, increasing its future responsiveness to Glutamate
 - e.g. New AMPA receptors form, or old ones are made even more responsive
 - e.g. Some NMDA Receptors are changed into the easier-to-stimulate AMPA receptors
 - e.g. Activates enzymes in cell that break protein “bridges” that structure dendritic spines, splitting them to form **new dendritic branches** (increased surface area) lined with AMPA Receptors

Other structural changes include...

- **Pre-Synaptic**: - **Retrograde Messengers** (e.g. Nitrous oxide) from Post-Syn cell that prolong NT release
 - **Perforation** = division, expansion of “Active Zone” of Pre-Syn by out-growth from surface of Post-Synaptic cell
- **Genetic**: Activity can turn on transcription of DNA to RNA, then translation of RNA into proteins
 - e.g. Such proteins may change #, size, & distribution of NT vesicles, or other relevant metabolic processes in cell
- **Neurogenesis**: Rare in NS overall, but common in **Hippocampus**, esp re temporal-based and spatial learning

MEMORY- The active process of retrieval of something learned

Different types of memory appear to be mediated by different areas of the brain

- **Spatial** - **Hippocampus**: Recall of specific locations, spatial judgments of familiarity
- **Procedural** - **Cerebellum & Striatum** (Basal Ganglia): Motor Skill, How to do it (peck a target, ride a bike)
- **Declarative** - **Hippocampus & Mediodorsal Thalamus**: Episodic (personal history), Semantic/Associative (facts)

Spatial Memory: _

- Hippocampus esp active in rat learning a maze, developing a **Cognitive Map** of its environment
 - **Place Cells** in Hippocampus are differentially active when rat is in different, familiar locations
- Birds that cache thousands of seeds & must remember locations during winter (e.g. Clark’s Nutcracker)
 - have much larger Hippocampus than non-caching relatives (e.g. Scrub Jay)
 - Also larger in Humans who have extensive spatial experience (e.g. taxi drivers, bushmen)
- Humans show more activity (e.g. via PET scans) in Hippocampus while answering questions that depend on spatial information (e.g. re: locations in or routes through city) than nonspatial (e.g. Who’s Who)
- Plus, damage to Hippocampus impairs formation/use of spatial memory (e.g. ability to navigate, to map)

Procedural Memory:

- **Cerebellum** damage interferes with performance in Cond **A (Procedural Memory)** but not Cond B,
while Hippocampus damage interferes with perf in Cond **B (Declarative/Spatial Memory)** but not on Cond A
- Also implicated in some cases of **Classical Conditioning** involving Procedural Memory
 - e.g. Rabbits: Tone (Conditioned Stim) + Puff of air at eye (Neg Reinf) =>Blink; Later Tone Alone=>Blink

- Damage to (or temporary suppression e.g. via cooling of) **Lateral Interpositus nucleus (LIP)** of Cerebellum
=> Even with extensive conditioning, rabbits never learn Tone + Blink association
- Damage to or suppression of **Red Nucleus** of Midbrain's Tegmentum (i.e. Motor site that LIP projects to)
=> Rabbit not show blink response until suppression wears off, then does show response
- So learned/stored in Cerebellum but not expressed without Midbrain participation
- **Striatum** (Caudate Nucleus & Putamen, input circuits of Basal Ganglia) also implicated in Procedural Memory
- **NMDA-Antagonist** injected into Striatum interferes with rat's recall of Cued Procedures
- As discussed previously, Basal Ganglia involved in selecting/integrating/ordering motor activity

Declarative Memory

- For cued **facts** or past **personal episodes**, perceptual input first processed by cortical sensory areas, then to "**Medial Temporal Lobe**" (Hippocampus & associated structures), then to **Mediodorsal Thalamus**, then to **Prefrontal Cortex** ("**Working Memory**") interconnected to other higher cortical areas
- **Hippocampus** not just involved in spatial memory, but also in consolidating & retrieving declarative memories
e.g. Match-To-Sample task (Shown Sample stimulus, subject picks which of 2 Alternatives matches Sample)
- Train rat on MTS until proficient; After rat learns task, lesion its **Hippocampus**
- Test w/novel stimuli, must apply "**rule**" (i.e. Pick alt that matches sample); Performance impaired
- **Mediodorsal Thalamus** (like Hippo.) damage more likely to impact declarative than procedural memory
- e.g. Connections from Thalamus to Prefrontal Cortex appear damaged in **Korsakoff's Syndrome**
- Chronic alcoholism >> vitamin **B1 (Thiamine)** deficiency, required for cells to metabolize glucose
- Anterograde amnesia (cannot form new memories) & Confabulation (make up stories based on current cues)
- **H.M.** Famous epilepsy patient had Hippocampus, Amygdala & some Temporal Cortex removed
- Personality & IQ fairly intact, but suffered severe Anterograde Amnesia (+ some Retro for time near trauma)
- Couldn't remember people he'd just met, page he'd just read; parents moved, couldn't find his way home
- Could be taught a simple associative task (e.g. word + photo), so Working Memory still OK
- But could not recall it 15 min later - So Hippocampus may act to **consolidate** some types of memories
- W/repetitive training, learned new skills (e.g. "Tower of Hanoi" puzzle) but not recall having learned them
- So **H.M.** shows Long-Term Procedural (Skill) Memory but not Declarative (Fact) Memory
- **Amygdala** itself also plays critical role in **Consolidation** of temporary associations into Long-Term Memories
- Block NMDA receptors in Basolateral Amygdala, prevent learning; But after learning, blocker has no effect !
- Emotion facilitates memory formation (Although extremes of emotion or stress can impair learning/memory)
- e.g. Arbitrary list of words easy to forget, except few "taboo" words that evoked emotional response

NOTE! Such memories **not stored** in Hippo. or Thalamus, but areas are necessary to consolidate/retrieve those memories

- e.g. Memory for familiar faces disrupted by damage to **Fusiform Gyrus** (of IT Cortex) = **Prosopagnosia**
- e.g. Memory for well-learned words, voices disrupted by damage to **Dorsal Temporal Cortex**
- e.g. Memory for activity in praxic space (within hands' reach, from egocentric view) in **Posterior Parietal**